SQA__Week4_Task

Internee Name: Mahzaib Fatima

DevelopersHub Corporation

Software Quality Assurance (SQA) – Weeks 4 to 6 Project Tasks

Advanced SQA, Performance Testing, Security Testing & QA Documentation

Submission Deadline: 28th April, 2026 Submission

Platform: Google Classroom

Performance Testing Report

Project Title: E-Commerce Website Load Test (Home, Book, Cart Pages)

Objective

The goal of this performance test was to:

- Simulate 750 concurrent users accessing three key pages of an e-commerce website (Home page, Book Page, Cart Page)
- Measure response times, throughput, and error rates under normal load
- Identify if the website becomes slow or unstable when multiple users access it at the same time
- Find bottlenecks (slow pages or high response times) that could harm user experience

Test Environment

Component	Details
Testing Tool	Apache JMeter (GUI mode)
Operating System	Windows / Linux (assumed)
Java Version	Java 8 or higher

Component	Details
Network	Local / Internal test environment
Target Application	E-commerce website (Home, Book, Cart pages)
Test Duration	Approximately 1-2 minutes (750 samples collected)

Listeners used: Mention in performance Test plan

- Aggregate Report
- Graph Results
- Summary Report

Load Test Scenario

Parameter	Value
Total Users (Threads)	750 (250 per page)
Pages Tested	Home Page, Book Page, Cart Page
Samples per Page	250
Total Samples	750
Ramp-Up Period	Not explicitly defined
Loop Count	1 (each user executed the request once)
Think Time	None (
Error Threshold	0%

Observations

4.1 Key Findings

Metric	Value	Observation
Overall Average Response Time	2483 ms (2.48 seconds)	Slightly slow but acceptable for a simple e-commerce site
Median Response Time	2110 ms (2.11 seconds)	Half of users experienced <2.1 sec – this is good
Minimum Response Time	334 ms (Cart Page)	Very fast under ideal conditions
Maximum Response Time	13007 ms (13 seconds!)	Critical issue – One user on Book Page waited 13 seconds
Error Rate	0.00%	Perfect – no request failed
Throughput	14.0 requests/sec	Server handled 14 users per second smoothly

4.3 Percentile Analysis (Overall)

Percentile	Value	Meaning
50% (Median)	2110 ms	Half of users waited ≤2.1 sec
90%	4341 ms	10% of users waited >4.3 sec
95%	5475 ms	5% of users waited >5.5 sec

Percentile	Value	Meaning
99%	7665 ms	1% of users waited >7.7 sec

Graphs & Results: mention in performance test plan

Graph Results (Visual Interpretation)

- X-axis: Samples 1 to 750
- Y-axis: Response time (0 ms to 4952 ms scale)
- Observed pattern: Response times fluctuate throughout the test
- Latest sample: 654 ms (fast)
- Deviation: 1527 ms (high variability)

Conclusion & Recommendations

Conclusion

The application performed well under the given load. It handled multiple users efficiently with acceptable response times and no major failures.

Recommendations

- Optimize slow pages to reduce response time
- Improve server capacity for higher loads
- Monitor performance regularly
- Conduct testing with higher users (e.g. 200+) for future analysis

FINAL STATEMENT SUMMARY

A load test was conducted on an e-commerce website's Home, Book, and Cart pages using Apache JMeter with 750 total samples. The system achieved 0% errors and handled 14 requests per second successfully. However, the average response time (2.48 seconds) exceeded the ideal threshold of 2 seconds, and the 95th percentile (5.47 seconds) indicates poor experience for 1 in 20 users. A critical 13-second spike was observed on the Book Page. Recommendations include optimizing the Home Page, investigating the 13-second outlier,

implementing caching, and improving database queries. The system passes reliability but fails speed targets.